ITA0605 - Machine Learning
List of Lab Exercises

8. Write a program to implement Linear Regression (LR) algorithm in python.

Aim

To implement the **Linear Regression (LR)** algorithm in Python and predict the output values based on the input data.

Algorithm

1. Import the required Python libraries.
2. Load the dataset.
3. Select the input feature (`X`) and target (`y`).
4. Split the dataset into training and testing sets.
5. Create a Linear Regression model.
6. Train the model using the training data.
7. Predict the output for the test data.
8. Calculate and display the model accuracy (R^2 score).

Dataset 1:

Program Code

```
from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

data = load_diabetes()

X = data.data

y = data.target
```

```
X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)
model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

score = model.score(X_test, y_test)

print("Predicted Values:")

print(y_pred[:10])

print("R2 Score:", score)
```

Output

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Load dataset
data = load_diabetes()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy (R² Score)
score = model.score(X_test, y_test)

print("Predicted Values:")
print(y_pred[:10])
print("R2 Score:", score)
```

```
Predicted Values:
[139.5475584  179.51720835 134.03875572 291.41702925 123.78965872
  92.1723465  258.23238899 181.33732057  90.22411311 108.63375858]
R2 Score: 0.4526027629719195
```

Dataset 2:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
X = np.array([
    [2, 70, 60],
    [3, 75, 65],
    [4, 80, 70],
    [5, 82, 75],
    [6, 85, 80],
    [7, 88, 85],
    [8, 90, 90],
    [9, 92, 94],
    [10, 95, 96],
    [11, 97, 98]
])
```

y = np.array([55, 60, 68, 74, 80, 86, 91, 95, 98, 100])
model = LinearRegression()
model.fit(X, y)
new_student = np.array([[7, 90, 88]])
prediction = model.predict(new_student)
print("Student Marks Prediction")
print("------------------------")
print("Study Hours: 7")
print("Attendance: 90%")
print("Assignment Score: 88")
print("Predicted Final Marks:", round(prediction[0], 2))

Output:

```
# Display result
print("Student Marks Prediction")
print("------------------------")
print("Study Hours: 7")
print("Attendance: 90%")
print("Assignment Score: 88")
print("Predicted Final Marks:", round(prediction[0], 2))
```

```
...  Student Marks Prediction
     ------------------------
     Study Hours: 7
     Attendance: 90%
     Assignment Score: 88
     Predicted Final Marks: 90.0
```

Dataset 3:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
X = np.array([
    [1, 15, 60],
    [2, 15, 65],
    [3, 16, 70],
    [4, 16, 75],
    [5, 17, 80],
    [6, 17, 82],
    [7, 18, 85],
    [8, 18, 88],
    [9, 18, 90],
    [10, 19, 94]
])
y = np.array([3.5, 4.2, 5.1, 6.0, 7.2, 8.0, 9.1, 10.0, 11.2, 12.5])
model = LinearRegression()

model.fit(X, y)
new_employee = np.array([[7, 18, 86]])
prediction = model.predict(new_employee)
```

```
print("Employee Salary Prediction")
print("--------------------------")
print("Experience: 7 years")
print("Education Level: 18 years")
print("Skill Score: 86")
print("Predicted Salary: ₹", round(prediction[0], 2), "Lakhs")
```

Output:

```
    print("--------------------------")
    print("Experience: 7 years")
    print("Education Level: 18 years")
    print("Skill Score: 86")
    print("Predicted Salary: ₹", round(prediction[0], 2), "Lakhs")

... Employee Salary Prediction
    --------------------------
    Experience: 7 years
    Education Level: 18 years
    Skill Score: 86
    Predicted Salary: ₹ 9.15 Lakhs
```

Dataset 4:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
X = np.array([
    [1.0, 900, 60],
    [1.2, 950, 70],
    [1.4, 1050, 80],
    [1.6, 1150, 90],
    [1.8, 1250, 100],
    [2.0, 1350, 110],
    [2.2, 1450, 120],
    [2.4, 1550, 130],
    [2.6, 1650, 140],
    [2.8, 1750, 150]
])
y = np.array([24, 22, 20, 18, 16, 15, 13, 12, 11, 10])
model = LinearRegression()
model.fit(X, y)
new_car = np.array([[1.5, 1100, 85]])
prediction = model.predict(new_car)
print("Car Fuel Efficiency Prediction")
print("------------------------------")
print("Engine Capacity: 1.5 L")
print("Vehicle Weight: 1100 kg")
print("Speed: 85 km/h")
print("Predicted Mileage:", round(prediction[0], 2), "km/L")
```

Output:

```
# Display result
print("Car Fuel Efficiency Prediction")
print("------------------------------")
print("Engine Capacity: 1.5 L")
print("Vehicle Weight: 1100 kg")
print("Speed: 85 km/h")
print("Predicted Mileage:", round(prediction[0], 2), "km/L")
```

```
... Car Fuel Efficiency Prediction
    ------------------------------
    Engine Capacity: 1.5 L
    Vehicle Weight: 1100 kg
    Speed: 85 km/h
    Predicted Mileage: 18.97 km/L
```

Dataset 5:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
X = np.array([
    [10, 2, 120],
    [15, 3, 150],
    [20, 4, 180],
    [25, 5, 210],
    [30, 6, 240],
    [35, 7, 260],
    [40, 8, 280],
    [45, 9, 300],
    [50, 10, 320],
    [55, 11, 340]
])
y = np.array([3.5, 4.5, 5.8, 7.0, 8.2, 9.5, 10.8, 12.0, 13.5, 15.0])
model = LinearRegression()

model.fit(X, y)
new_sales = np.array([[38, 7, 270]])

prediction = model.predict(new_sales)
print("Sales Prediction")
print("----------------")
print("Advertising Cost: ₹38 Thousand")
print("Salespersons: 7")
print("Store Visits: 270")
print("Predicted Monthly Sales: ₹", round(prediction[0], 2), "Lakhs")
```

Output:

```
print("Salespersons: 7")
print("Store Visits: 270")
print("Predicted Monthly Sales: ₹", round(prediction[0], 2), "Lakhs")
```

```
Sales Prediction
----------------
Advertising Cost: ₹38 Thousand
Salespersons: 7
Store Visits: 270
Predicted Monthly Sales: ₹ 10.28 Lakhs
```

Result:

Thus, the Linear Regression algorithm was successfully implemented for House Price Prediction, Student Marks Prediction, Employee Salary Prediction, Car Fuel Efficiency Prediction, and Sales Prediction. The model was trained using the given datasets and successfully predicted the target value for each new sample.